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IN ASTROPHYSICS AND SPACE SCIENCE

MASTER'S THESIS

RECONSTRUCTING PHOTOMETRIC LIGHTCURVES
OF ACTIVE GALACTIC NUCLEI USING NEURAL
STOCHASTIC FLOWS

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УНИВЕРЗИТЕТ У БЕОГРАДУ
МАТЕМАТИЧКИ ФАКУЛТЕТ



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Реконструкција фотометријских светлосних
кривих активних галактичких језгара
применом стохастичких Neural Flow модела

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1 Background

Compared to ordinary galaxies, some have extraordinarily bright central regions, which can sometimes even outshine the host-galaxy emission by many orders of magnitude; these galaxies are called *active galaxies* and their nuclei are referred to as *active galactic nuclei* (AGN). The physical processes necessary to create such high luminosities ($L = 10^{43-48} \text{ erg s}^{-1}$; e.g., Lusso et al. 2012) within such a small volume were a subject of debate for years, but now are widely accepted to be connected to accretion onto the galaxies' central black holes (BHs; Salpeter 1964; Zeldovich 1964). These accreting objects have masses $M_{BH} \sim 10^{6-10} M_{\odot}$ (e.g., Peterson and Horne 2004) and an Eddington ratio exceeding $\lambda_{Edd}^1 \approx 10^{-5}$. This threshold, chosen somewhat arbitrarily, excludes galactic nuclei with no significant accretion activity, such as that of the Milky Way, while still covering their full range of observed luminosities (Netzer 2015).

The “unified model” (Antonucci 1993; Urry and Padovani 1995) discussed in Section 1.1, posits the black hole at the center of the AGN, surrounded by several distinct components, including an accretion disk, gas and dust clouds, and, in some cases, a relativistic jet (Figure 1.4). Each of these regions contributes emission into different wavelength regimes, resulting in these objects having detectable emission across the whole electromagnetic spectrum, as illustrated by the spectral energy distribution (SED) in Figure 1.1.

The gravitational energy released as the circumnuclear material falls inwards and loses angular momentum forms the accretion flow around the black hole, which acts as the primary source of AGN continuum emission. The disk gets heated up via viscous heating (Peterson 1993), producing a thermal spectrum that has a blackbody temperature peaking in the ultra violet (UV; $\approx 10 - 400 \text{ nm}$) for a “typical” AGN (i.e., with $M_{BH} \approx 10^8 M_{\odot}$; $\lambda_{Edd} \approx 0.1$); this is often referred to as the “big blue bump” (Harrison 2014; Figure 1.1)

The thermal disk photons, also called *seed photons*, are Comptonized by highly energetic electrons that are located at the *corona* (§1.1.2) to form the X-ray power-law spectrum observed in Figure 1.1. These high energy photons can also reflect off the accretion disk and/or the *torus* (§1.1.4), producing the additional hard X-ray “reflection” component. In a majority of AGN X-ray spectra, there is an additional “soft excess” (Singh et al. 1985; Arnaud et al. 1985) at energies less than $\sim 2 \text{ keV}$. This emission is above what would be expected from simple accretion disk models and its origin is still a topic of discussion. Two proposed models (which could be simultaneously true) to explain this soft excess are: relativistically blurred ionized reflection from the

¹The ratio L/L_{Edd} , where L is the bolometric luminosity and $L_{Edd} = 1.5 \times 10^{38} M_{BH}/M_{\odot} \text{ erg s}^{-1}$ is the Eddington luminosity for a solar composition gas. This is the maximum isotropic luminosity a body can achieve while maintaining balance between radiation pressure and gravitational force.

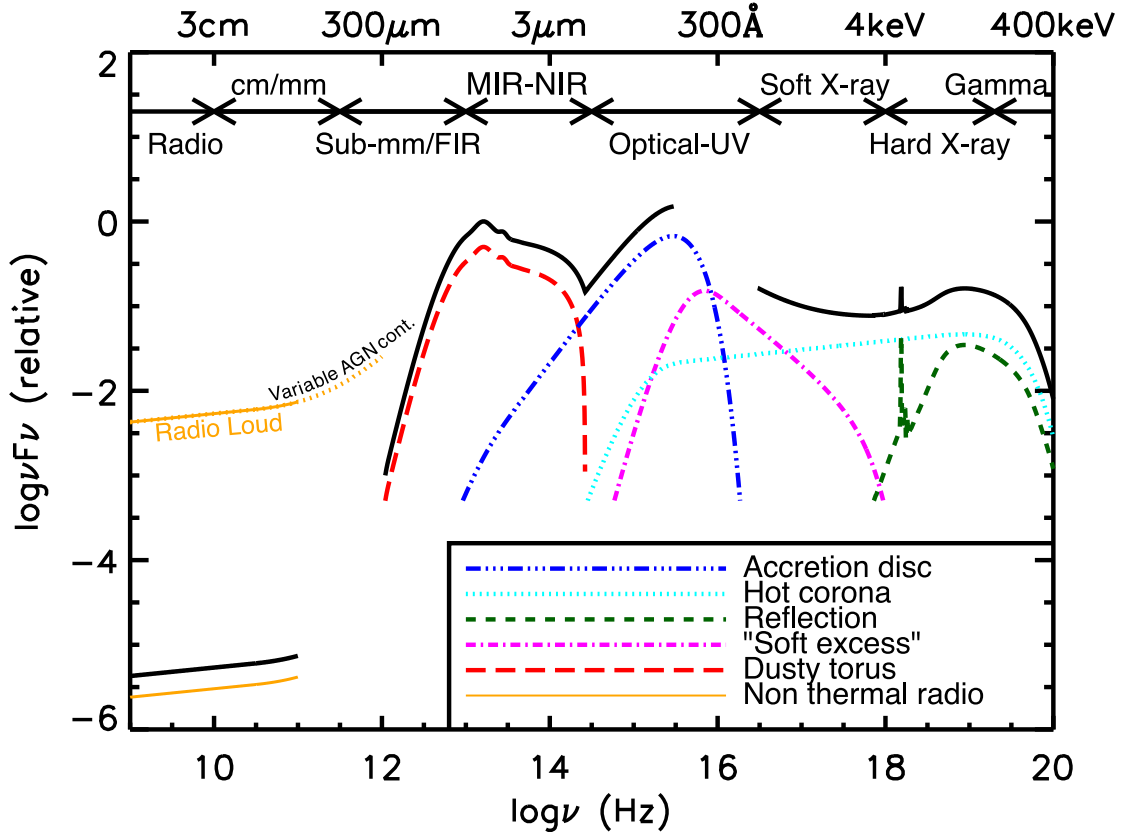


Figure 1.1: AGN SED from radio frequencies to hard X-ray, loosely based on the observed SEDs of radio-quiet quasars (e.g., Elvis et al. 1994; Richards et al. 2006). The black solid curve represents the total emission and the colored curves (shifted down for clarity) represent the individual constituents. The intrinsic shape of the SED in the mm-far infrared (FIR) regime is uncertain; however, it is credited to have a minimal contribution (to an overall galaxy SED) compared to star formation (SF), except in the most intrinsically luminous quasars and powerful jetted AGN. It should be noted that the relative contribution of each of the components can vary dramatically for different AGN types. Adapted from Harrison (2014).

inner accretion disc (e.g., Crummy et al. 2006), or Comptonization in a “warm corona” located near the surface of the accretion flow (e.g., Ballantyne and Xiang 2022).

The infrared (IR; $\approx 1 - 1000 \mu\text{m}$) band, is mostly sensitive to the reprocessed emission of the accretion disk by the torus. Although the shape of this IR emission varies from source to source, the peak of this dusty component is usually around $\lambda \approx 20 - 50 \mu\text{m}$ and falls steeply at longer wavelengths (Harrison 2014). In the sub-mm/far-infrared (FIR), or sometimes even in the mid-infrared (MIR), stellar heated emission from cooler dust dominates over AGN heated emission (Lutz 2014).

AGN cover a wide range of radio luminosities, which can differ by several orders of magnitude between radio-quiet (RQ) and radio-loud (RL) sources, now more aptly referred to as non-jetted and jetted, respectively ² (orange lines in Figure 1.1). The latter type makes up only a minority of the AGN population (e.g., Padovani 2011) with objects that present highly collimated, relativistic radio *jets* (§1.1.5) and extended radio structures such as lobes and hotspots, that together emit strong non-thermal radiation and, occasionally, a γ -ray jet. On the other hand, the spectrum of non-jetted objects is dominated by thermal emission in most bands and the origin

²While RL/RQ has been the historical terminology, radio-quiet sources are not truly radio-silent but rather “radio-faint” and they are, however, γ -ray quiet (Padovani 2017).

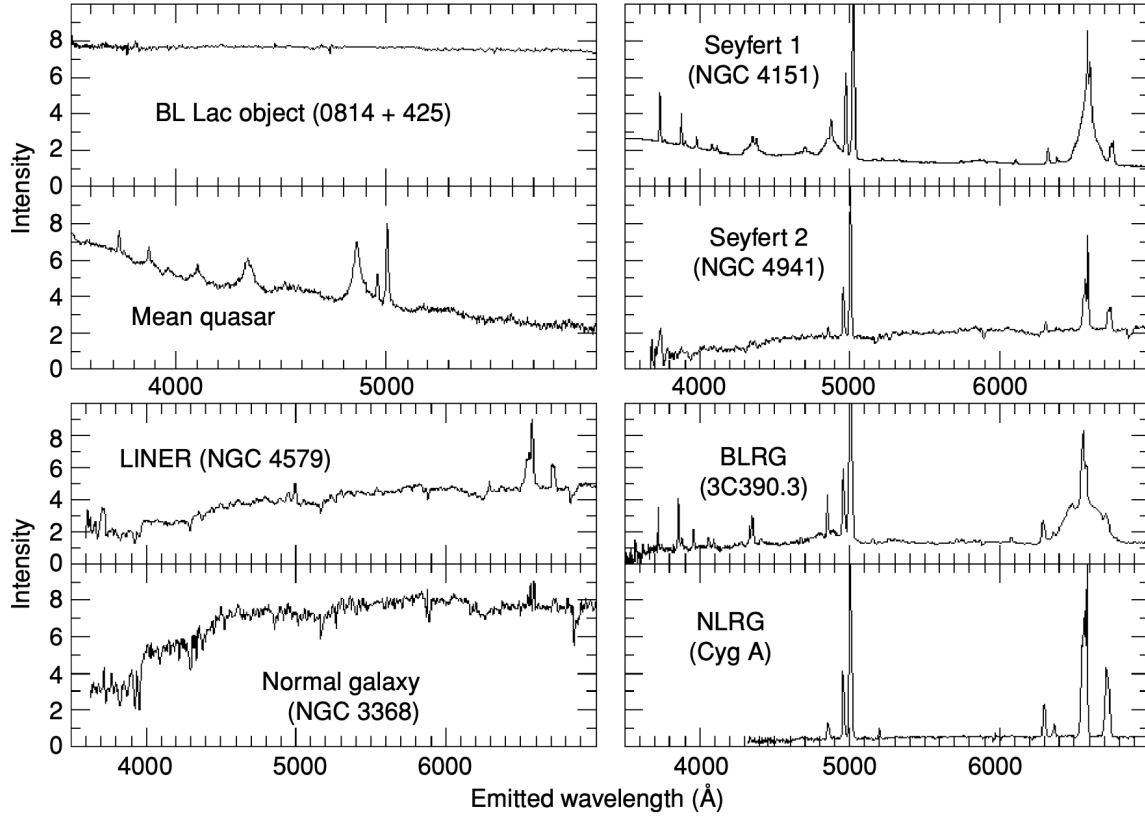


Figure 1.2: Optical spectra of representative AGN classes, from the (essentially featureless) BL Lac objects (Top left) through the radio galaxies and QSOs (Seward and Charles 2010). Note the presence of both broad and narrow emission lines in quasars, Seyfert 1 galaxies and broad-line radio galaxies (BLRG), while Seyfert 2 galaxies, low-ionisation nuclear emission region (LINER) galaxies and narrow-line radio galaxies (NLRG) only show narrow permitted and forbidden lines. A normal galaxy spectrum (Bottom left) is shown for comparison.

of their radio emission has not been clearly established. Some possibilities include coronal activity or synchrotron radiation from particles accelerated in shocks (see e.g., Laor and Behar 2008; Ishibashi and Courvoisier 2011).

1.1 Unified Model of AGN

Because different wavebands probe different physical elements, AGN selected at particular wavelengths often exhibit distinct properties, leading to a rich phenomenological classification (Figure 1.2). Reality, however, is much simpler because now we know that most of these classes differ from each other only by a small number of parameters, such as: orientation, accretion rate, the presence of jets, and possibly the host galaxy and environment (see Padovani et al. (2017) and references there-in). This suggests that the numerous brands do not necessarily correspond to fundamentally different physical objects, but instead motivate a unifying framework capable of explaining their apparent diversity.

This unification idea arose shortly after the discovery of *quasars* (QSOs³), when the observed

³The term quasar historically stands for “quasi-stellar radio source”; throughout this thesis it is used more broadly to refer to very luminous AGN.

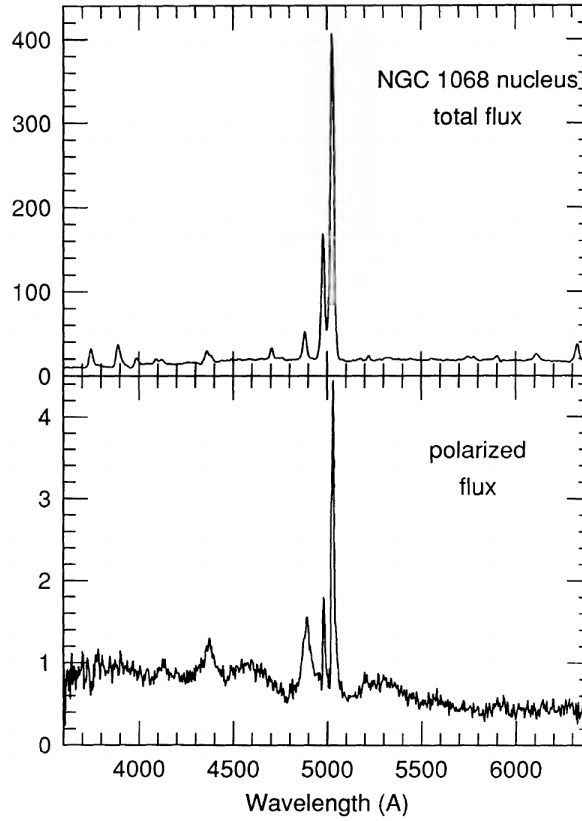


Figure 1.3: Optical spectra of the nucleus of NGC 1068 (Miller, Goodrich, and Mathews 1991). The total spectrum (Top) shows the narrow emission lines expected for Type 2 AGN, while the polarized flux spectrum (Bottom) reveals the broad permitted emission lines characteristic of unobscured Type 1 AGN.

variety of AGN prompted questions about the role of the observer’s orientation relative to the AGN axis. Blandford and Rees (1978) proposed that BL Lac objects were radio galaxies viewed down the axis of a relativistic jet, where *relativistic beaming* (§1.1.5) would cause the non-thermal continuum to appear very bright and the emission lines weak in comparison. The same object, viewed from the side, would have normal emission-line equivalent widths, and the radio structure would be dominated by the extended lobes rather than the core (Shields 2004).

Further evidence came from spectropolarimetry, the measurement of the polarization state of the incoming light from a source as a function of wavelength. Seyfert galaxies (Seyfert 1943) are primarily divided into two types based on their optical spectra: Type 1 objects exhibit both broad permitted emission lines such as $H\alpha$ and $H\beta$ ($\text{FWHM} > 1000 \text{ km s}^{-1}$) and narrow forbidden lines such as $[\text{O III}]$ and $[\text{Ne III}]$ ($\text{FWHM} < 1000 \text{ km s}^{-1}$), while Type 2 objects show only the narrow component. Crucially, both types share the same high- and low-ionization narrow forbidden lines with similar line ratios, suggesting they are powered by the same intrinsic engine.

Antonucci and Miller (1985) confirmed this by analyzing the polarized flux of NGC 1068, a Seyfert 2 galaxy⁴, resembled that of a Seyfert 1 spectrum, as can be seen in Figure 1.3. This was interpreted as the broad-line region (BLR; §1.1.3) and the central continuum source being blocked from direct view by a dusty torus, while free electrons above the nucleus scatter a fraction of the nuclear light towards the observer, polarizing it in the process. The broad lines had gone unnoticed in direct light because this scattered component was too faint relative to the

⁴It should be noted that a subset of Type 2 AGN ($\sim 30\%$), sometimes referred to as “true Type 2” objects, show no detectable broad lines even in polarized light and little X-ray absorption (see Netzer 2015 and references there-in).

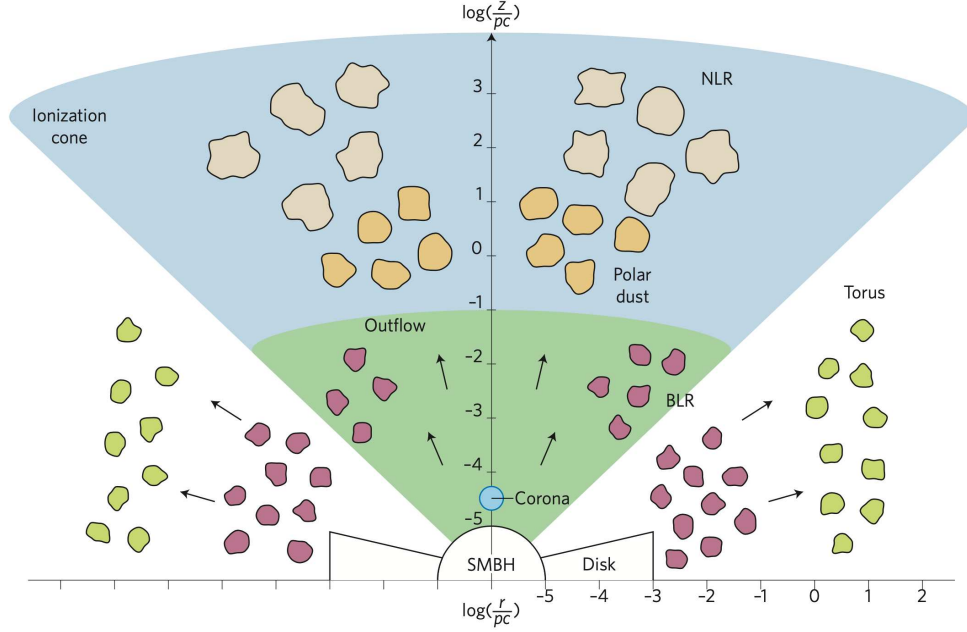


Figure 1.4: Overview of the innermost regions around a non-jetted accreting supermassive black hole (SMBH). Typical scales along the polar and equatorial directions (for nearby AGN) are shown. Different colors depict regions with different densities or compositions. Taken from Ramos Almeida and Ricci 2017.

strong narrow-line emission from the narrow-line region (NLR; §1.1.6), which extends beyond the obscuring torus and is therefore visible regardless of orientation.

Multiple efforts such as those recounted above (e.g., Antonucci 1984; Orr and Browne 1982) were fundamental to the development of the “favoured” unified model of AGN introduced in this section. Whilst it has been successful in homogenizing many kinds of active nuclei, recent works point to the conclusion that a picture based primarily on orientation and obscuring material is incomplete (e.g., Bianchi, Maiolino, and Risaliti 2012). Nonetheless, the major aspects of the model are sufficient for the purposes of this thesis; in the sections below, the primary components are explained briefly, while additional features such as polar dust and large-scale outflows (also visible in Figure 1.4) are discussed shortly when relevant but are not treated as separate subsections.

1.1.1 Accretion Disk

The standard framework of the optically thick, geometrically thin accretion disk (Shakura and Sunyaev 1973) described above applies to SMBHs with moderate to high Eddington ratios ($10^{-2} \lesssim \lambda_{Edd} \lesssim 1$; Ricci 2026), where the accretion flow is considered radiatively efficient. This scheme describes the disk as a composition of Keplerian rings, nested and coupled to each other by viscous forces, likely driven by turbulence and magnetic instabilities, the magneto-rotational instability (MRI; Balbus and Hawley 1998) being the most widely quoted. The outer, slower ring brakes the inner faster one, transporting the angular momentum outwards and allowing material to spiral inward. Friction between the rings produces heat that is dissipated as blackbody emission with a characteristic maximum temperature of $T \sim 10^5$ K for the “typical” AGN described above (Netzer 2013), yielding the multi-temperature spectrum depicted in Figure 1.1. The continuum emission of the accretion disk varies with time depending on the dynamics of the infalling matter (see Chapter 2).

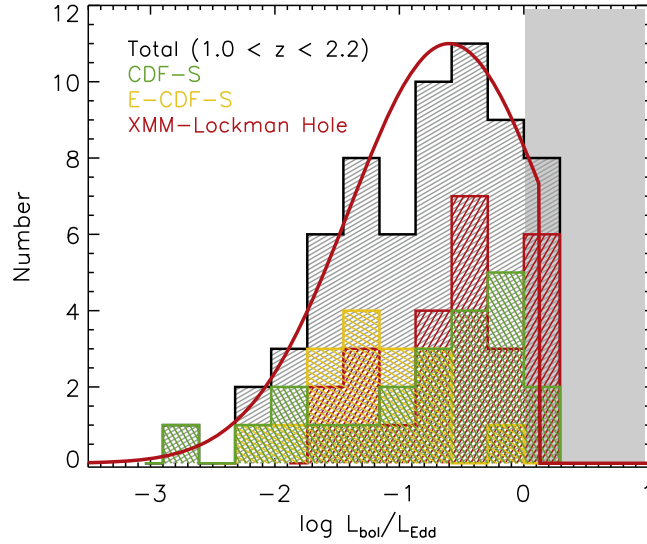


Figure 1.5: Eddington ratio distribution of broad-line AGNs at $1.0 < z < 2.2$. Different X-ray surveys are represented as different color histograms, and the black histogram pictures all the combined distributions. The gray shade illustrates the Eddington limit ($\lambda_{Edd} = 1$) and the red solid line corresponds to a log-normal fit with a peak of $\log L/L_{Edd} = -0.6$ (Suh et al. 2015).

Most QSOs in the local universe are thought to accrete through thin disks, yet contrasting accretion scenarios are possible depending on the Eddington ratio (Figure 1.5). At $\lambda_{Edd} \lesssim 10^{-2}$, the disk becomes optically thin and with an Advection Dominated Accretion Flow (ADAF, Narayan and Yi 1994), radiating weakly and with minimal optical/UV emission. Conversely, at $\lambda_{Edd} \gtrsim 1$, theoretical works have predicted that the SMBHs accrete rapidly through “slim” (or “thick”) disks, showing very different SEDs than their slower counterparts and powerful outflows (e.g., Kubota and Done 2018; Giustini and Proga 2019).

1.1.2 Corona

Numerical simulations of magnetohydrodynamical accretion disks have shown that the dissipation of the magnetic energy built up by the MRI naturally produces a hot, active corona located within a few gravitational radii ($R_g = GM/c^2$) of the accretion disk (Balbus and Hawley 1998; Miller and Stone 2000; Fabian et al. 2009). This compact region of ionized gas reaches temperatures of $T \sim 10^9$ K (Kara and García 2025), and its rapid variability (see Chapter 2) is consistent with such a small emitting region. The corona contributes approximately 5% of the AGN bolometric output (Ricci 2026), a fraction that decreases with increasing λ_{Edd} , with lower λ_{Edd} systems showing stronger X-ray emission relative to the disk (Padovani et al. 2017).

1.1.3 Broad-line Region

The broad-line region consists of high density gas clouds ($\gtrsim 10^9 \text{ cm}^{-3}$) predominantly in Keplerian motion at luminosity-dependent distances of $0.01 - 1$ pc from the central black hole (Netzer 2015), well within the *dust sublimation radius* (see §1.1.4). The BLR is therefore dust free, emitting broad permitted and semi-forbidden emission lines with velocity widths ranging from 1,000 to 20,000 km s^{-1} (Ricci 2026), wideness caused by *Doppler broadening*⁵. Ionizing photons from the

⁵Doppler broadening refers to the superposition of emission from clouds moving at different directions/velocities along the line of sight produces a single broadened line whose width reflects the velocity dispersion of the gas.

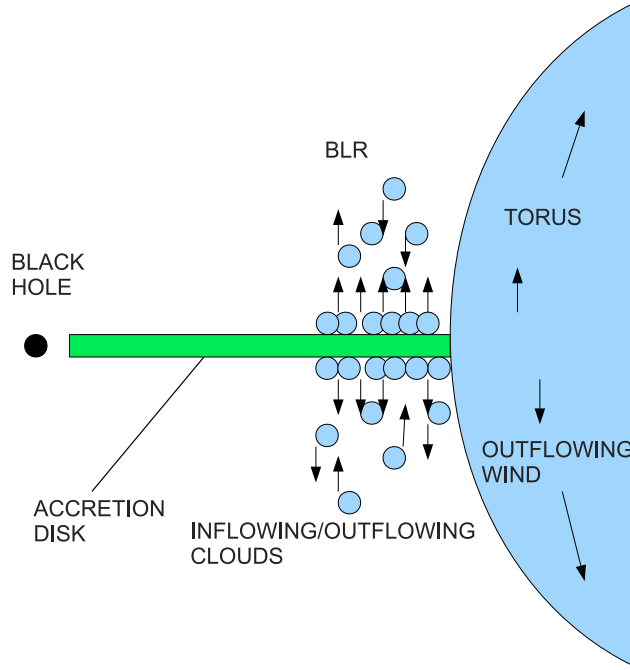


Figure 1.6: Schematic of the BLR geometry and its location relative to the accretion disk and dusty torus. The figure also illustrates the failed wind scenario of Czerny and Hryniewicz (2011): dust forms in the atmosphere of the inner accretion disk at radii where $T_{eff} < T_{sub}$, is launched as a wind, but then is suppressed by the radiation from the central source and partially falls back, giving rise to the inflowing and outflowing clouds that make up the BLR.

accretion disk and corona illuminate these clouds, which subsequently exhibit ionization stratification with respect to radius, with higher ionization species located closer to the central source (Netzer 2013). The physical origin of the BLR remains an open question, some of the models presented in the literature include radiation pressure driven disk winds (Murray et al. 1995) and failed dust-driven wind from the disk surface (Figure 1.6; Czerny and Hryniewicz 2011). Notably, the distance of the BLR from the central source scales with the AGN luminosity as $R_{BLR} \propto L^{1/2}$ (Kaspi et al. 2000), a relationship that is fundamental in reverberation mapping studies (§2.1.1).

1.1.4 Dusty Torus

As posed previously, the presence of an asymmetric “dusty obscurer” has successfully unified apparently different AGN classes. When observed edge-on (i.e., at high inclination angles), this torus⁶ obscures the innermost regions of the AGN and hides the broad emission lines; conversely, when viewed face-on (at lower inclination angles), there is a clear line of sight to the BLR and accretion disk (Seyfert 1 vs Seyfert 2 in Figure 1.2). This structure is conceptualized as an optically thick structure surrounding the central engine at scales larger than that of the BLR and luminosity dependent dimensions ranging from 0.1 – 10 pc (Netzer 2015). Its inner boundary coincides with the dust sublimation radius (R_{sub}), beyond which the dust can survive the radiation from the central engine. This radius corresponds to the distance at which the dust temperature reaches the sublimation threshold ($T_{sub} \sim 1400 - 1800$ K for silicate and graphite grains respectively; e.g., Barvainis 1987; Netzer 2015).

⁶For convenience, we keep referring to this obscuring matter as the torus despite the fact that it could have a different (and more complex) geometrical distribution (e.g., Nenkova et al. 2008; Stalevski et al. 2012).

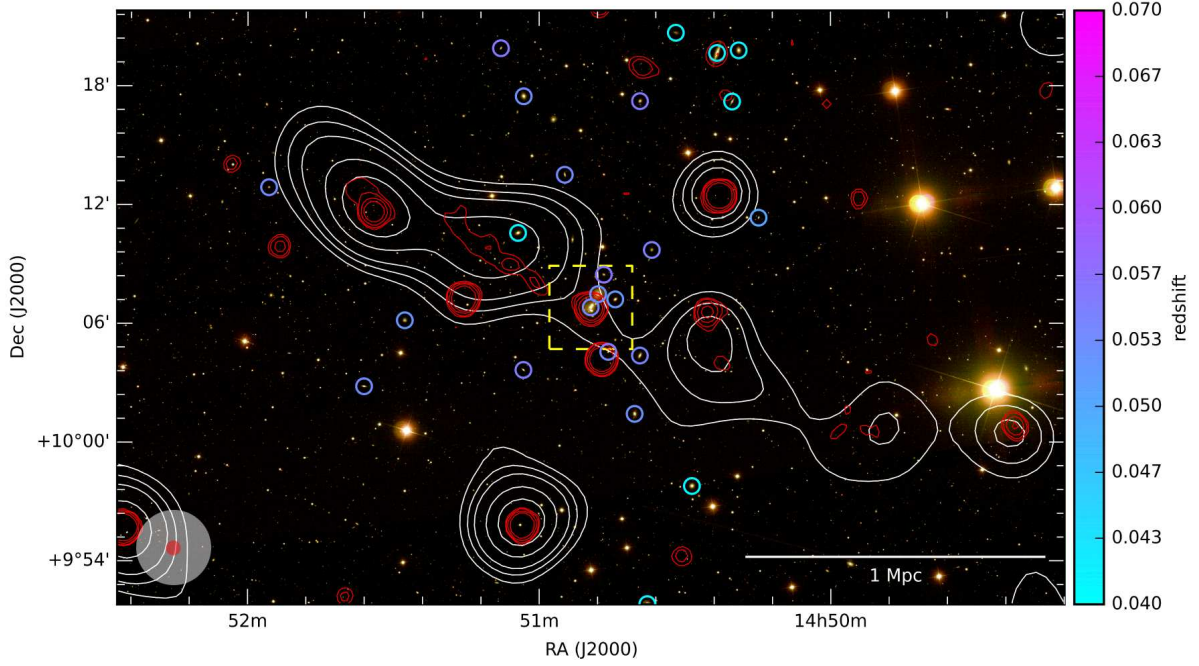


Figure 1.7: SDSS optical image (composite from bands g , r and i) of the galaxy group UGC 9555 overlaid with NVSS contours (red) and LOFAR M5SS radio contours (white) revealing the size of the giant radio galaxy (GRG). The white bar indicates a physical scale of 1 Mpc, illustrating how AGN jets can extend well beyond the boundaries of their host galaxies. Adapted from Clarke et al. 2017.

In addition to being responsible for the IR reprocessed emission on the AGN SED (Figure 1.1), the torus acts as a gas and dust reservoir that regulates the material available for accretion onto the black hole. Its geometry, believed to contain a central opening, also collimates the AGN radiation into bi-conical ionization cones along the polar direction (Pogge 1988), which illuminate the NLR (§1.1.6). Furthermore, AGN drive large scale winds and outflows along these directions, powered by radiation and magnetic pressure from the central engine, which can interact with and regulate the host galaxy interstellar medium (Ricci 2026).

1.1.5 Relativistic Jets

Jets are thought to be powered by the extraction of rotational energy from the SMBH via strong electromagnetic fields (Blandford and Znajek 1977), and can extend to Mpc scales from their host galaxy (Figure 1.7; Clarke et al. 2017).

Their launch point coincides with the location of the corona, in the immediate vicinity of the accreting object, and their apparent velocities range from mildly to highly relativistic motion. The broadband emission of jets is characterised by a double-peaked SED: the low-energy peak, spanning radio to optical wavelengths, arises from synchrotron radiation of relativistic electrons spiralling in the jet magnetic field, while the high-energy peak, from X-ray to γ -ray energies, is thought to result from inverse-Compton scattering of soft photons from the BLR and accretion disk by the same relativistic electrons (Netzer 2013).

Their observed properties depend strongly on orientation because relativistic beaming could make an approaching jet appear significantly brighter than a receding one. This effect, known as *Doppler boosting*, arises when the jet plasma has bulk relativistic motion that concentrates the emitted radiation into a narrow cone in the direction of motion, amplifying the observed flux and variability for jets viewed close to face-on and explaining the extreme characteristics observed

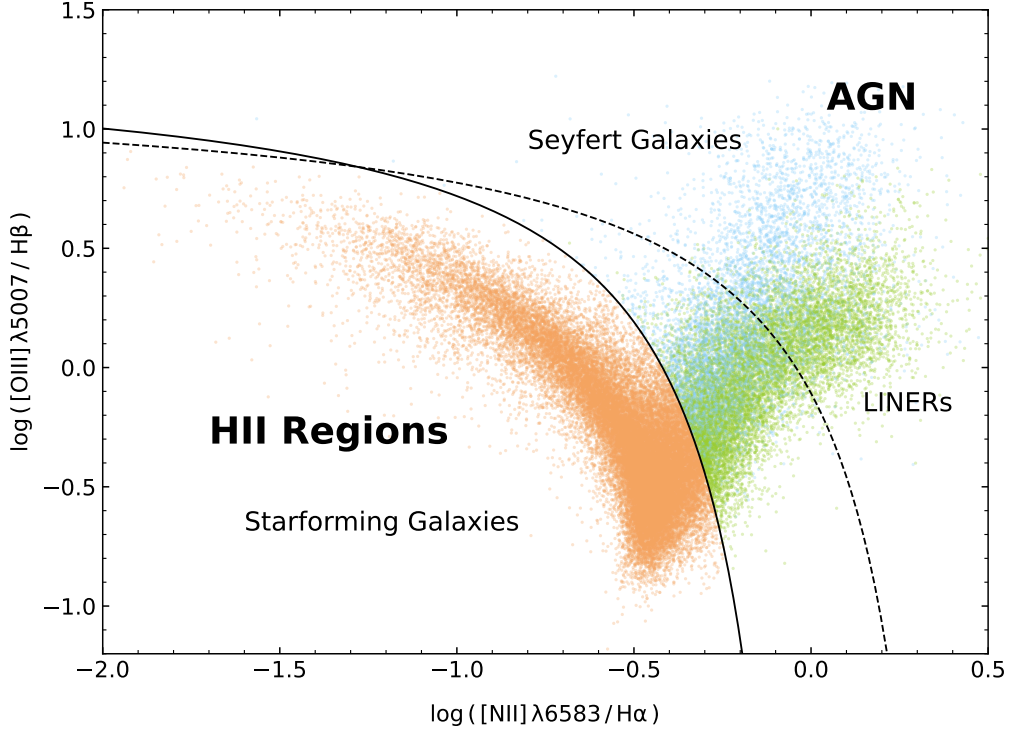


Figure 1.8: Select galaxy objects from the Sloan Digital Sky Survey (SDSS) catalog shown in the BPT diagram. It plots the intensity of the [OIII] $\lambda 5007$ forbidden transition relative to that of the $H\beta$ line versus the ratio of [NII] $\lambda 6583$ to $H\alpha$ on logarithmic axes. Using this to infer ionization states and abundances of the emitting medium enables us to reconstruct the origin of the dominant radiation in the galactic host, which has led to various approaches for the classification of these objects. Depicted is the theoretical delimiter by Kewley et al. (2001) as well as the empirical line by Kauffmann et al. (2003) to distinguish predominantly starforming galaxies from AGN. Additionally, the Schawinski et al. (2007) boundary sorts AGN into efficiently accreting Seyfert and inefficient LINER subtypes. Courtesy of Fritz A. Agildere.

on blazars (Urry and Padovani 1995). Jets are also thought to play a role in galaxy evolution by injecting energy into the surrounding environment, a process referred to as radio-mode feedback (Fabian 2012).

1.1.6 Narrow-line Region

This assymetric region is extended enough to be resolved optically (e.g., Fischer et al. 2013; Polack et al. 2024), spanning hundreds or even thousands parsecs along the ionization cones (Figure 1.4). The NLR, just as the BLR, is photoionized by the central engine and exhibits optical emission lines, kinematically broadened with typical widths of $10^2\text{--}3 \text{ km s}^{-1}$ (Bennert 2005), with most lines falling within the $350 - 500 \text{ km s}^{-1}$ range (Ricci 2026).

Compared to the BLR, the NLR has a lower electron density ($\sim 10^{2-3} \text{ cm}^{-3}$), allowing forbidden transitions to occur without being collisionally suppressed. The presence of these forbidden lines, together with the high ionization state of the gas, can trace AGN activity and makes the NLR emission diagnostically distinct from that of star-forming regions, a separation commonly visualised using emission line ratio diagrams such as the Baldwin-Phillips-Terlevich (BPT) diagram (Baldwin, Phillips, and Terlevich 1981) (Figure 1.8).

2 Variability of AGN

The first quasar discovered, 3C 273 (Edge et al. 1959), had been documented in photographic plates at least as early as 1887 (Slavcheva-Mihova et al. 2013). This emblematic radio-loud object showed variable emission in the optical and radio bands on timescales from days to years (Smith and Hoffleit 1963; Dent 1965), indicating that the size of the emitting region spanned only a few light-days in size. This compactness, coupled with the finding of its distant redshift (Schmidt 1963), was very difficult to explain by dynamics that operate in normal galaxies, and variability thus became one of the key arguments that the enormous luminosities of quasars had to originate from an exceptionally compact and energetic central engine (Salpeter 1964; Zeldovich 1964).

Now we know that variability in all wavelengths is an ubiquitous property of AGN, and so, its studies have been essential in understanding the physics of their nuclear regions, which, in general, cannot be resolved by current telescopes. This had been strictly true until the observations of Sgr A* and M87 (The Event Horizon Telescope Collaboration et al. 2019) and, more recently, with the GRAVITY interferometer on the VLT resolving the outer BLR in 3C 273 (Gravity Collaboration et al. 2018); still, these surveys are limited to massive sources in the very nearby universe. For the vast majority of AGN, variability studies therefore rely on the *light curve*, a time series recording how the brightness of the AGN, measured within a given wavelength band, varies over a monitoring period. The time scales, spectral changes, and the correlations and delays between variations extracted from these light curves provide crucial information on the nature and location of these components and on their interdependencies (Ulrich et al. 1997).

2.1 Physical Origin and Timescales

Early campaigns quickly revealed that AGN vary much more rapidly in X-rays than in the optical (Winkler and White 1975), sometimes on timescales as short as a few minutes (McHardy et al. 1981), and that variability at different wavelengths is correlated (Ulrich et al. 1997). This stochastic behaviour can generally be explained by the *propagating fluctuations* theory (Lyubarskii 1997). Local fluctuations in the disk's surface density, presumably driven by the MRI (§ 1.1.1), occur at all radii and propagate inwards at the local viscous timescale¹. As these fluctuations travel, they modulate the local mass accretion rate at progressively smaller radii, coupling the variability of the outer disk to that of the hotter inner regions and corona, where the shorter local viscous

¹The viscous timescale, $t_{\text{visc}} \sim r^2/\nu$, where ν is the kinematic viscosity, indicates the time for material to drift radially through the disk and increases with radius.

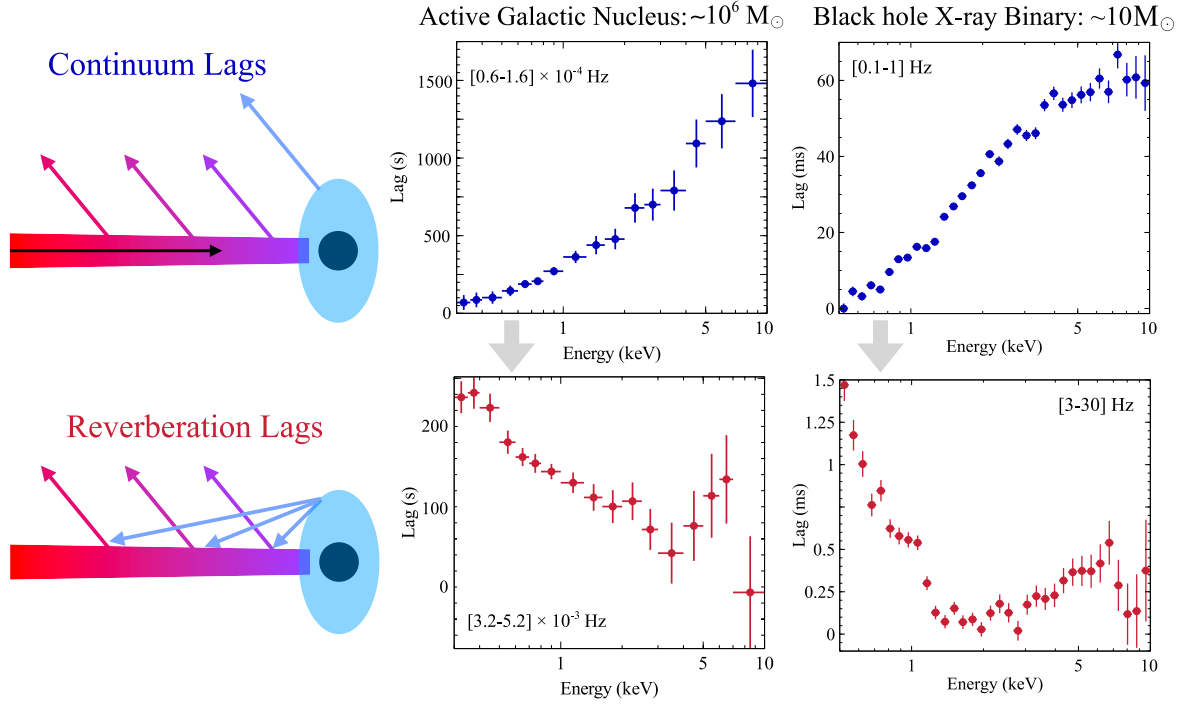


Figure 2.1: Frequency-resolved X-ray time lags as a function of energy for an AGN ($\sim 10^6 M_\odot$, left) and a black hole X-ray binary ($\sim 10 M_\odot$, right), illustrating the universality of the two lag regimes. At low Fourier frequencies (top, blue), hard X-rays lag soft ones, consistent with inward-propagating accretion fluctuations. At high Fourier frequencies (bottom, red), soft X-rays lag hard ones, consistent with disk reverberation. Taken from Kara and García 2025.

timescale drives faster fluctuations in the emitted luminosity (Uttley et al. 2023). The X-ray variability produced in the innermost regions (i.e., in the corona) then irradiates the outer accretion disk and is reprocessed, driving the optical and UV variability observed at longer wavelengths with a characteristic time delay (Padovani et al. 2017), a *reverberation*, as discussed below.

Beyond this persistent stochastic variability, more extreme variability phenomena also exist, including *changing-look* AGN, which undergo dramatic changes in their broad emission lines and continuum on timescales of months to years, possibly driven by rapid changes in accretion rate or obscuration (LaMassa et al. 2015), as well as quasi-periodic eruptions (QPEs) and tidal disruption events (TDEs), they represent transient phenomena superimposed on the underlying stochastic variability and are not discussed further here.

Both propagating fluctuation lags and reverberation lags are simultaneously present in the data, but dominate at different temporal frequencies and have opposite energy dependences (Figure 2.1). Fourier analysis allows the separation of both lag types into two temporal regimes: at low frequencies ($\nu \lesssim 10^{-5}$ Hz for a $10^6 M_\odot$ black hole, or equivalently timescales $\gtrsim 1$ day), hard X-ray variations lag soft ones as accretion fluctuations propagate inward; at high frequencies ($\nu \gtrsim 10^{-3}$ Hz, timescales of minutes), soft X-rays lag hard ones as the disk responds to the driving corona (Kara and García 2025). Simple cross-correlation averages over all timescales and can obscure this distinction, making Fourier analysis essential for disentangling the two contributions.

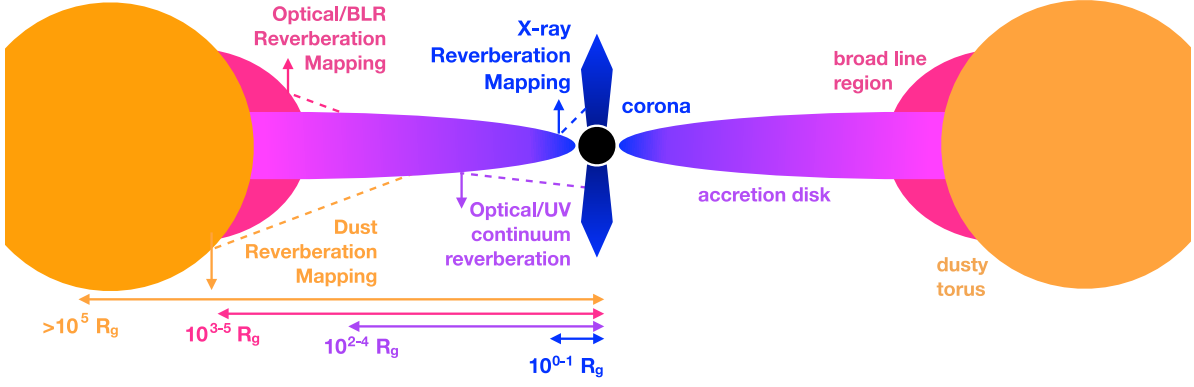


Figure 2.2: Cross-sectional schematic of an AGN illustrating the four main types of reverberation mapping and their corresponding radial scales from the central black hole. Taken from Cackett et al. 2021.

2.1.1 Reverberation Mapping

Reverberation mapping (RM; Blandford and McKee 1982) exploits the fact that any region illuminated by the central engine responds to its variability with a characteristic lag $\tau = R/c$, where R is the light travel distance to the responding region, providing a direct physical scale (Peterson 2006).

Formally, the observed emission-line light curve $L_l(t)$ is related to the driving continuum $L_c(t)$ through a convolution with the *transfer function* $\Psi(\tau)$:

$$y(t) = \int_{-\infty}^{\infty} \Psi(\tau) y_{drive}(t - \tau) d\tau, \quad (2.1)$$

where $\Psi(\tau)$ encodes the geometry and kinematics of the responding region (Netzer 2013). The centroid of $\Psi(\tau)$ gives the mean lag $\langle \tau \rangle$, and hence the characteristic size of the responding region. The characteristic timescales and size scales probed by different types of reverberation are summarised in Table 2.1 and Figure 2.2.

Table 2.1: Order of magnitude size scales and their equivalent light-crossing times for the different regions of an AGN. Adapted from Cackett et al. 2021.

Region	Size scale (R_g)	Light-crossing time	
		$10^7 M_\odot$	$10^9 M_\odot$
X-ray corona	10	500 s	14 hr
UV/optical disk	10^2 – 10^4	0.06–6 days	6–600 days
BLR	10^3 – 10^5	0.6–60 days	60–6000 days
Dusty torus	$> 10^5$	> 60 days	> 6000 days

In the optical/UV, the most established application is BLR reverberation mapping, which measures the lag between continuum and broad emission line variations; since both the continuum and the broad lines are observed to vary in a correlated fashion, the measured lag directly yields the BLR size via $R_{BLR} = c\tau$ (Peterson 2006). As introduced in Section 1.1.3, this size scales with AGN luminosity as $R_{BLR} \propto L^{1/2}$ (Kaspi et al. 2000), a relationship empirically calibrated through RM campaigns and used to estimate black hole masses via the virial theorem². Contin-

² $M_{BH} = f R_{BLR} \Delta V^2 / G$, where ΔV is the emission line width and f is a geometric factor (Peterson 2006).

uum reverberation mapping instead measures the wavelength-dependent lags between different optical/UV bands, probing the accretion disk size and temperature profile. Under the standard irradiated disk model, the lag at wavelength λ scales as (Netzer 2022):

$$\tau_{\lambda} \propto \lambda^{4/3} \left[\frac{M_{\text{BH}} \dot{M}}{(3 + f_{\text{irr}})} \right]^{1/3}, \quad (2.2)$$

where $\dot{M}$ is the accretion rate and f_{irr} parameterizes the fraction of X-ray flux absorbed by the disk. This $\lambda^{4/3}$ scaling arises from the temperature profile of the standard thin disk ($T \propto r^{-3/4}$), with each wavelength probing emission from a characteristic disk radius. Observed continuum lags, however, are often larger than predicted by this simple model (Cackett et al. 2021), partly because diffuse BLR emission contributes to the broadband continuum at larger radii than the disk itself, biasing the inferred lags towards longer values (Korista and Goad 2001; Netzer 2022). Separating these contributions remains an active area of research, with microlensing studies offering an independent constraint on the disk size (Cackett et al. 2021).

2.2 Variability Characterization

Variability has been studied under a variety of methods, starting from simple statistical quantities tracing the amplitude and timescales of the observed variability to more sophisticated approaches in both the time and frequency domains (Paolillo and Papadakis 2025). In general, these statistics attempt to measure the absolute or fractional variance of the light curve over some characteristic timescale Δt , or express the degree of correlation between different timescales or photometric bands (Ulrich et al. 1997; De Cicco et al. 2025).

The distribution of variability power over temporal frequencies (or, equivalently, 1/timescale) is quantified by the *power spectral density* (PSD), defined as the squared amplitude of the flux (e.g., Uttley et al. 2002). AGN PSDs typically follow a “red-noise” power law, $P(\nu) \propto \nu^{-\alpha}$ with $\alpha \sim 1$ at low frequencies, steepening to $\alpha \gtrsim 2^3$ above a characteristic break frequency ν_b , consequently displaying larger amplitude variations on longer timescales. This break timescale scales with black hole mass and accretion rate (McHardy et al. 2006), with smaller black holes varying more rapidly than larger ones for the same observing duration (Figure 2.3). The anticorrelation can be interpreted as a consequence of the fact that the most luminous AGN are likely to host larger mass black holes (if all quasars accrete at similar Eddington rates), in which case, their variability should be diluted by the larger emitting region (Paolillo and Papadakis 2025).

The PSD also depends strongly on the observing waveband, with optical PSDs showing steeper slopes and lower normalization than X-ray PSDs, consistent with the propagating fluctuations picture where outer disk regions contribute power at lower frequencies and inner regions at higher ones (Kara and García 2025).

A complementary time-domain characterization is provided by the *structure function* (SF), which quantifies the mean variability amplitude as a function of time lag Δt ; on short timescales the SF rises as a power law, flattening above a characteristic damping timescale τ that encodes the same physical information as the PSD break frequency (Kozłowski 2016).

Variability-based methods are generally more effective than SED-based methods at identifying low-mass and low-luminosity AGN, since variability provides a cleaner signature of nuclear activity against the host galaxy background (YuEzTao2025). Nonetheless, deriving the PSD/SF

³ $\alpha \sim 2$ corresponds to a *Damped random walk* that breaks to flat slope at long timescales.

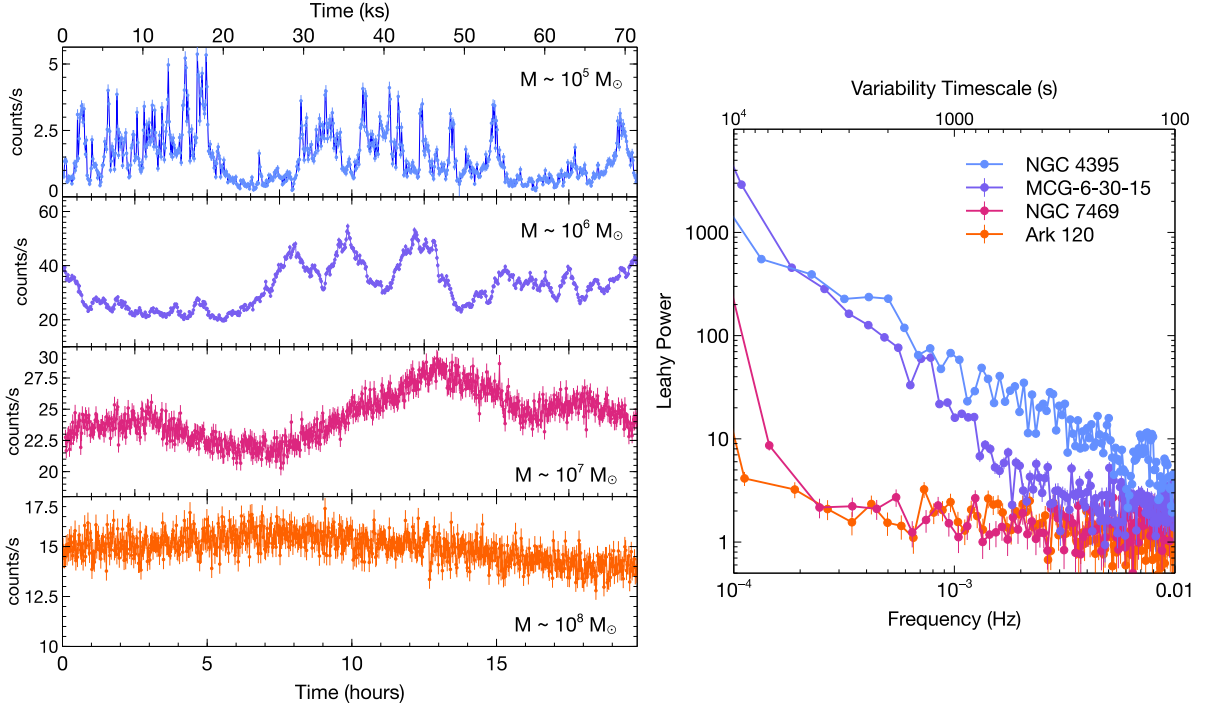


Figure 2.3: Left: X-ray light curves of four Seyfert 1 AGN with black hole masses spanning 10^5 – $10^8 M_\odot$, observed with XMM-Newton over 70 ks. Smaller black holes vary more rapidly, with the $10^5 M_\odot$ AGN (NGC 4395, top) showing variability on timescales of tens of seconds, while the $10^8 M_\odot$ AGN (Ark 120, bottom) varies on timescales of hours. Right: The corresponding PSDs, showing the red-noise power-law shape characteristic of AGN X-ray variability. Taken from Kara and García 2025.

features and their uncertainties from an irregularly sampled light curve is very challenging, specially for stochastic light curves such as those of AGN; thereon, high quality light curves are indispensable to enable robust characterization of the UV/optical variability and interband lags (Kelly et al. 2014).

3 Simulation Framework

This chapter describes the framework used to generate realistic AGN light curves with which the reconstruction method of Chapter 4 is trained and evaluated. The intrinsic variability is modeled using the EzTAO Gaussian process library (Yu and Richards 2022), which is then observed through an irregular, gappy cadence with added photometric noise, mimicking the sampling of a ground-based monitoring campaign.

3.1 The Vera C. Rubin Observatory LSST

3.2 Damped Random Walk

The simplest stochastic model consistent with the observed AGN PSD properties is the *damped random walk* (DRW), also known as a first-order continuous autoregressive process, CAR(1), or Ornstein-Uhlenbeck process (Kelly, Bechtold, and Siemiginowska 2009). It is described by the stochastic differential equation:

$$df(t) = -\frac{f(t)}{\tau_{\text{DRW}}} dt + \sigma_{\text{DRW}} \sqrt{dt} \epsilon(t), \quad (3.1)$$

where τ_{DRW} is the damping (or characteristic) timescale, σ_{DRW} is the asymptotic variability amplitude, and $\epsilon(t)$ is a white noise process.

$$X_{t+\Delta t} | X_t \sim \mathcal{N}(\mu + e^{-\Delta t/\tau} (X_t - \mu), \sigma^2 (1 - e^{-2\Delta t/\tau})), \quad (3.2)$$

The DRW has an exponential autocorrelation function:

$$\text{ACF}(\Delta t) = \exp\left(-\frac{|\Delta t|}{\tau_{\text{DRW}}}\right), \quad (3.3)$$

corresponding to a PSD of the form $P(\nu) \propto \nu^{-2}$ at high frequencies, flattening to white noise below the break frequency $\nu_b = 1/(2\pi\tau_{\text{DRW}})$. In the structure function representation, the DRW produces a power-law rise with slope $\gamma = 0.5$ on short timescales ($\Delta t \ll \tau_{\text{DRW}}$), flattening to a constant value set by σ_{DRW} above the characteristic timescale (Figure 3.1; Kozłowski 2016). The DRW has been shown to provide a good statistical description of optical AGN light curves (Kelly, Bechtold, and Siemiginowska 2009; Kozłowski et al. 2010; MacLeod et al. 2010), with τ_{DRW} found

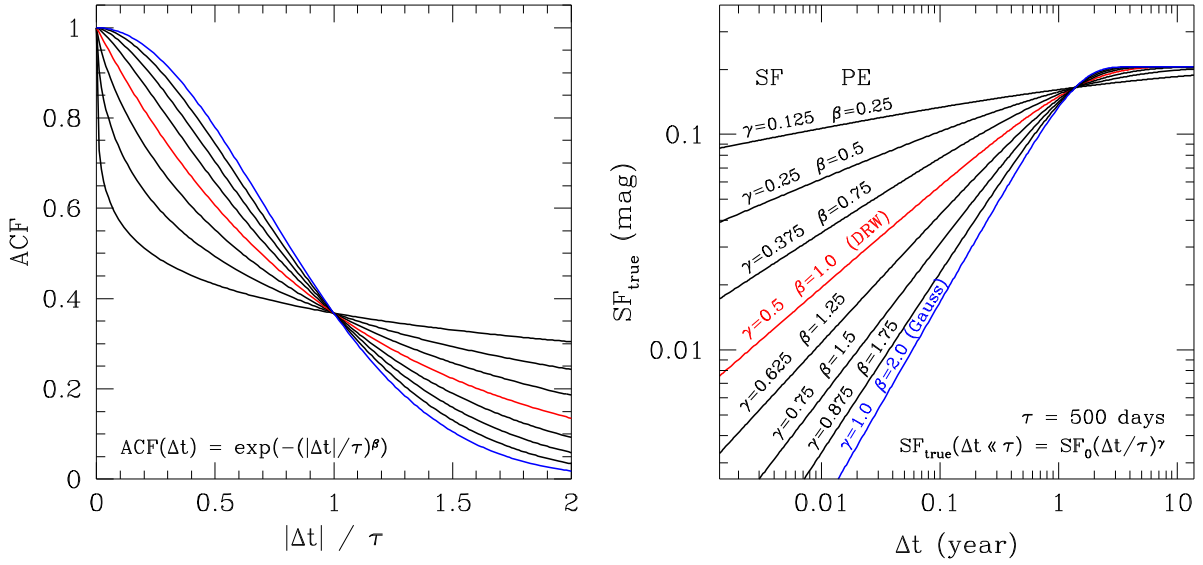


Figure 3.1: Left: Exponential autocorrelation functions $ACF(\Delta t) = \exp(-(|\Delta t|/\tau)^\beta)$ for different values of β . Right: The corresponding structure functions. The DRW model ($\beta = 1$, red) produces a SF with slope $\gamma = 0.5$ on timescales $\Delta t \ll \tau$, flattening above the characteristic damping timescale. Adapted from Kozłowski 2016.

to correlate with black hole mass and luminosity, consistent with the physical picture developed in Chapter 2.

The DRW is the simplest member of the broader family of continuous autoregressive moving average (CARMA) models (Kelly et al. 2014), corresponding to the special case CARMA(1, 0). More flexible CARMA(p , q) processes can reproduce a wider range of PSD shapes; in particular, the CARMA(2, 1) process, also known as the damped harmonic oscillator (DHO), has been shown to provide superior fits to observed AGN light curves compared to the DRW (Yu et al. 2025). The simulation framework described here assumes a DRW process for the underlying variability, though the implementation supports more general CARMA kernels.

w

3.2.1 EzTao Implementation

3.2.2 Noise Model

Observational noise is simulated by adding Gaussian random values drawn from $\mathcal{N}(0, \sigma_{\text{LSST}}^2)$ to each flux measurement, where $\sigma_{\text{LSST}} = 0.01 \text{ mag}$ is the expected LSST single-visit photometric uncertainty¹.

3.2.3 Damping Timescale

Rather than sampling τ from an arbitrary range, we draw it from the distribution implied by the population of AGN black-hole masses. Burke et al. (2021) established a correlation between the

¹<https://rubinobservatory.org/for-scientists/rubin-101/key-numbers>

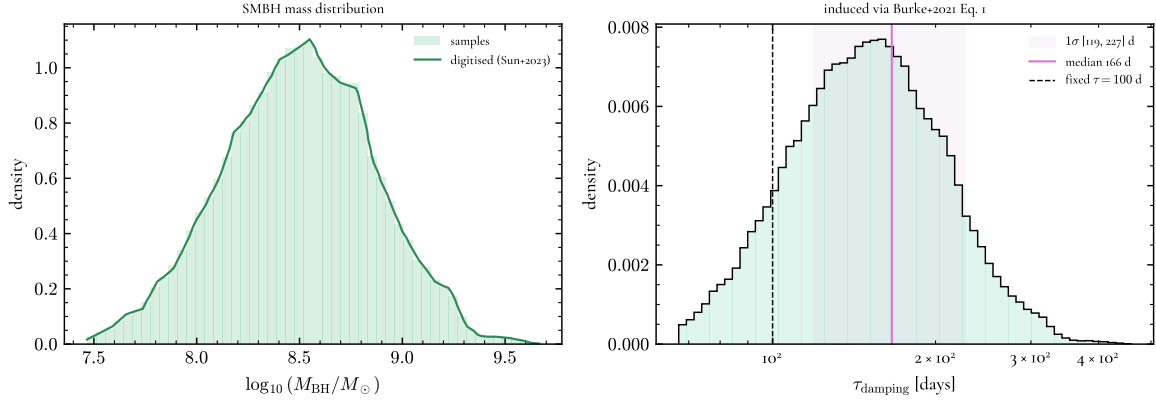


Figure 3.2: .

optical damping timescale and the black-hole mass,

$$\tau_{\text{damping}} = 107^{+11}_{-12} \text{ days} \left(\frac{M_{\text{BH}}}{10^8 M_{\odot}} \right)^{0.38^{+0.05}_{-0.04}}, \quad (3.4)$$

which we combine with the bias-corrected black-hole mass distribution of Sun (2023) for their preferred model. Each simulated curve draws a value of $\log M_{\text{BH}}$ from the **Sun2023QuasarMassDistDist** distribution by inverse-transform sampling, which is then mapped to a damping timescale through Equation 3.4. Sampling the full distribution, including its high-mass tail, yields a training set whose τ distribution reflects the physical quasar population rather than a uniform prior;

3.2.4 Sampling Strategy

The simulation supports two modes: a single-band driver light curve (**makeDriver**), which generates a single DRW realisation at the reference band parameters on a uniform time grid with step Δt up to a maximum baseline t_{max} ; and a multiband mode (**makeMultiband**), which generates correlated light curves in all requested bands simultaneously using the wavelength-scaled parameters. A sparse sampling mode (**makeSparse**) is also implemented to mimic realistic LSST observing cadences by subsampling the uniform grid with gaps and irregular spacing.

3.3 Simulation Regime

4 Models

This chapter describes the model used to reconstruct the gaps in the simulated light curves of Chapter 3. The intrinsic variability is a solution of a stochastic differential equation, and the natural object for gap filling is therefore the transition density $p(x_t | x_s)$ of that equation. For the damped random walk this density is known in closed form (Equation 3.2), but for more general processes it is not, and evaluating it requires a numerical solver run between every pair of observation times. Neural Stochastic Flows (**Kiyohara2025NSF**) circumvent this by learning the transition density directly as a conditional normalising flow, evaluable in a single network pass for an arbitrary time separation.

4.1 Neural Stochastic Flows

4.2 Training Objective

The flow and an auxiliary bridge model are trained jointly with the objective of **Kiyohara2025NSF**,

$$\mathcal{L}(\theta, \xi) = \underbrace{-\mathbb{E}_{\mathcal{D}} [\log p_{\theta}(x_{t_j} | x_{t_i})]}_{\mathcal{L}_{\text{NLL}}} + \lambda \underbrace{\mathcal{L}_{\text{flow}}(\theta, \xi)}_{\text{consistency}}, \quad (4.1)$$

comprising two terms with complementary roles. The first is a negative log-likelihood evaluated on *observed* transition pairs (x_{t_i}, x_{t_j}) ; it anchors the learned density to the data. The second enforces the Chapman–Kolmogorov consistency that a direct transition and a two-step transition through an intermediate time must agree,

$$p(x_k | x_i) = \int p(x_k | x_j) p(x_j | x_i) dx_j, \quad (4.2)$$

whose intractable intermediate marginal is handled by introducing a bridge model $q_{\xi}(x_j | x_i, x_k)$ as an auxiliary variational distribution, yielding a tractable bidirectional bound (**Kiyohara2025NSF**).

Both terms are required. The consistency term alone admits any self-consistent dynamics, of which the true process is only one; without the likelihood term nothing selects the correct transition, and a model so trained reproduces observed light curves only because the bridge is supplied with real endpoints at evaluation time. The likelihood term is evaluated over triplets (t_i, t_j, t_k) whose spans are drawn to cover the full baseline, so that transitions across long time

separations—including the direct $t_i \rightarrow t_k$ pair—are constrained directly by the data.

4.3 Data Normalisation

The flow conditions on the pair $(x, \Delta t)$, and is poorly conditioned if presented with $\Delta t \sim 1000$ and $x \sim 0.1$. The light curves are therefore normalised so that time is measured in units of the baseline, $t \rightarrow t/t_{\max}$, and flux in units of the driver amplitude, $y \rightarrow y/\sigma$, giving a normalised timescale $\tau_n = \tau/t_{\max}$ and unit variance. The normalization is applied consistently to the validation set using the scales computed from the training set.

4.4 Gap Filling

Reconstruction of a gap is performed with the bridge model, conditioned on the states at both bracketing observations. A distribution conditioned on both endpoints is, by construction, a stochastic bridge; the transition density alone, conditioned only on the earlier endpoint, would instead constitute a forecast, whose spread grows without bound rather than closing on the later endpoint. For a given gap the bridge is sampled on a dense interior grid, and the reconstruction is summarised by the pointwise median and central credible intervals of the resulting Monte Carlo ensemble.

4.5 Latent

4.6 Validation and Baseline

4.6.1 The Learned Transition

Because the driver is a damped random walk, its transition density is known analytically (Equation 3.2), and the learned density $p_\theta(x_t | x_s, \Delta t)$ can be compared against it directly. Sampling the learned transition at a range of separations and comparing the sample mean and standard deviation against the analytic $e^{-\Delta t/\tau} x_s$ and $\sigma\sqrt{1 - e^{-2\Delta t/\tau}}$ provides a direct test that the model has learned the underlying dynamics, and not merely a density consistent with the observed pairs (Figure ??). This validation is specific to the present setting: the benchmark systems of Kiyohara2025NSF possess no closed-form transition against which such a comparison could be made.

4.6.2 The Bayes-Optimal Baseline

As a reference we compare against a Gaussian process regression employing the true damped-random-walk kernel, $k(\Delta t) = \sigma^2 e^{-|\Delta t|/\tau}$, supplied with each curve’s own τ and σ . Since the simulated data are generated from exactly this process, the Gaussian process conditional distribution is the exact posterior over the unobserved flux, and its mean is the minimum-mean-square-error reconstruction. Given the true parameters, no method can produce a more accurate mean or a narrower calibrated interval; the baseline therefore represents a ceiling rather than a

competitor. The predictive mean and variance at prediction times t_* are

$$\mu_* = K_* (K + \sigma_{\text{phot}}^2 I)^{-1} y, \quad (4.3)$$

$$\Sigma_* = K_{**} - K_* (K + \sigma_{\text{phot}}^2 I)^{-1} K_*^\top, \quad (4.4)$$

where K , K_* and K_{**} are the kernel evaluated among the observations, between prediction and observation times, and among the prediction times respectively (**RasmussenWilliams2006GP**). The additive measurement term on the diagonal ensures that the reconstructed interval contracts to $\sim \sigma_{\text{phot}}$ at an observation rather than to zero, reflecting that an observation constrains the latent flux only up to the measurement error.

4.6.3 Evaluation Metrics

Reconstruction quality is quantified over the gap interiors only, excluding the pinned observation epochs at which both the reconstruction and the baseline are near-exact and the interval collapses to the noise level; their inclusion would inflate the apparent performance of both. For each validation curve we report the root-mean-square error of the median reconstruction and the empirical coverage of the credible intervals, the latter measuring calibration—the fraction of true values falling within a stated interval, which for a well-calibrated 68% interval should equal 0.68. Errors are additionally binned by gap width in units of τ , the variable that controls the information available from the bracketing observations, and pooled across the validation set so that the scaling of reconstruction error with gap size is characterised over the full population rather than for any single curve.

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